Mathis Goupillon

ENS Paris-Saclay Normalien (civil servant trainee), M2 MVA

Mathematics, Vision and Learning (ENS Paris-Saclay)

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Education

2026–2027 **Master 2 MVA - Mathematics, Vision and Learning**

École normale supérieure Paris-Saclay - Université Paris-Saclay

Master's program in applied mathematics for machine learning, computer vision, and data processing.

Planned coursework:

- Optimal transport for machine learning
- Convex optimization and applications in ML
- Graphs and Text in ML
- Probabilistic graphical model & Deep Generative models
- Foundation of LLM
- Computational statistics
- Geometric Data Analysis
- Representation Learning
- Reinforcement Learning
- Bayesian Machine Learning
- Deep Learning
- Kernel methods for ML
- Algorithms for speech and NLP
- Inverse problems and imaging: statistical and stochastic approaches
- Diffusion and multiscale models in AI

2025–2026 **Advanced Research Year in Artificial Intelligence**

École normale supérieure Paris-Saclay - Université Paris-Saclay

Research-oriented year in artificial intelligence, with a focus on deep learning, computer vision, and generative modeling. Two research internships: at Centre Borelli (ENS Paris-Saclay), and at MBZUAI (Abu Dhabi).

Planned Coursework :

- Foundation of AI & ML
- AI & ML for Time series and Signal modeling
- AI & ML for Network modeling
- AI & ML for Image and Video processing

2024–2025 **Master 1 EEA - Electrical and Digital Engineering Sciences**

École normale supérieure Paris-Saclay - Université Paris-Saclay

Interdisciplinary training in Machine learning, statistics, computer science, signal and image processing, artificial intelligence, and electronics.

Selected M1 grades: ***Class rank 1/27***

- Signal processing: **17.2/20**
- Information theory and source coding: **18.5/20**
- Image and signal processing: **18.1/20**
- Nonlinear control systems (Lie algebra): **16.3/20**
- Automatic control: **17.8/20**
- Reinforcement Learning: **17.7/20**
- Digital Signal Processing and Communications: **17.9/20**
- Embedded Systems: **17.1/20**
- Fundamentals of Communication Systems: **18.1/20**

2023–2024 **Dual Bachelor's Degree - Engineering Sciences (SAPHIRE program)**

École normale supérieure Paris-Saclay - Université Paris-Saclay

Training in Applied Sciences for Physics and Engineering, with a focus on research and teaching.

Selected grades: ***Overall Grade: 15.86***

- Computer science:
- Mathematics for modeling and simulation:
- Probability:
- System identification and optimization:
- Signal processing:

2021–2023 **CPGE - Intensive Preparatory Classes**

Lycée Jean Perrin, Marseille. Class rank 2/92

Intensive curriculum in mathematics, physics, and engineering sciences - admitted to ENS Paris-Saclay through the national competitive entrance examination.

Research Experience

April – August 2026 **Research Internship - Generative AI for Molecular Graphs**, *Mohamed bin Zayed University of Artificial Intelligence (Abu Dhabi)*, Ecole Polytechnique

Michalis Vazirgiannis - Roman Bresson

Latent Flow Matching for Molecular Graph Generation

- Developed a scalable generative framework for molecular graphs, combining graph Variational Autoencoders (VAEs) with latent Flow Matching, Variational Flow Matching (VFM), and classifier-based guidance to enhance molecular validity across three datasets of increasing complexity.
- Designed a regressor-based guidance strategy for property-constrained molecular generation, enabling precise control over target molecular properties while achieving particularly strong performance in conditional generation.
- Leveraged Optimal Transport to improve latent flow trajectories and explored unconstrained molecular generation as well as property-constrained generation.
- Achieved state-of-the-art or competitive performance against established graph generative models, demonstrating strong generation quality with significantly fewer sampling steps.
- Contributed to a research paper currently under review at ICLR 2027.

Nov – Jan 2025 - 2026 **Research Internship - Multimodal AI and Image Captioning**, *Centre Borelli (Paris)*
Prof. Miguel Colom Barco & Mehdi Boussaa
Vision-Language Models for Visual Geolocalizability Prediction
- Investigated multimodal AI approaches to assess image geolocalizability, aiming to identify geographically ambiguous images and improve the quality of visual geolocation training datasets.
- Leveraged state-of-the-art Vision-Language Models (VLMs), including Qwen3-VL and Moondream, to generate semantically rich image captions and extract geographic cues from visual content.
- Developed and evaluated a caption-based geolocalizability prediction pipeline, combining multimodal representations, CLIP embeddings, and Transformer-based cross-encoders to estimate image localization difficulty.
- Designed and evaluated a multimodal framework combining vision-language models, image captioning, and Transformer-based ranking to predict image geolocalizability and identify geographically ambiguous images.

May – July 2025 **Research Internship - Fetal Brain MRI Image Processing**, *Neuroscience Institut of La Timone (Marseille)*
Olivier Coulon & Guillaume Auzias
Fetal Brain MRI Image Processing
- Adapted and optimized the Morphologist pipeline (BrainVISA/Anatomist) for automated cortical sulci extraction from T2-weighted fetal MRI scans, addressing the challenges of fetal brain anatomy.
- Developed Python-based medical image processing algorithms and improved pipeline compatibility with fetal MRI data, supporting automated neuroanatomical analysis of brain development.

2024–2025 **Research Study Project (TER)**, *Université Paris-Saclay, Satie*
Antony Juton. Duration: **Throughout the academic year**
Deep Reinforcement Learning for Vision-Based Autonomous Driving
- Developed a deep reinforcement learning framework for autonomous racing, training a neural network-based driving policy using Proximal Policy Optimization (PPO) in the Webots simulator.
- Designed a visual perception pipeline using an Intel RealSense D435i RGB-D camera, processing depth images at 60 FPS to enable real-time, camera-based autonomous navigation without LiDAR.
- Optimized reward functions to balance speed and collision avoidance, and addressed the sim-to-real gap by calibrating camera geometry and vehicle dynamics for deployment on a physical robotic car.
- Successfully deployed the learned policy on an autonomous miniature racing car, achieving 3rd place out of approximately 25 teams in the inter-school CoVaPsy competition.

Publications and Articles

Submitted **M. Goupillon**, R. Bresson, K. Divriotis, M. Vazirgiannis. *Latent Flow Matching for Molecular Graph Generation* Submitted to *ICLR 2027*.[arXiv]

Academic Projects

2025–2026 **Few-Shot Image Classification with Meta-Learning**, *École normale supérieure Paris-Saclay*
Supervised by Gabriele Facciolo
- Implemented metric-based meta-learning with Prototypical Networks, comparing CNNs trained from scratch with pre-trained Vision Transformers (ViTs), achieving 97.3% accuracy on unseen classes in 5-way 5-shot classification.

2025-2026 **Unsupervised Learning for Parkinson's Disease Gait Analysis**, *École normale supérieure Paris-Saclay*
- Applied unsupervised learning and Gaussian Mixture Models (GMMs) to Parkinson's gait analysis, achieving 79% agreement with clinical labels.

2025–2026 **Transformers and Vision Transformers: A Mathematical and Architectural Tutorial**, *École normale supérieure Paris-Saclay*
Supervised by Nicolas Vayatis
Technical Tutorial – 17-page Academic Report
- Authored a technical tutorial on Transformer architectures, self-attention mechanisms, and Vision Transformers (ViTs).

Skills

Programming Python (PyTorch, NumPy, Pandas), C, MATLAB, Slurm, GPU/HPC computing
Software Git, VS Code, Linux, LaTeX.

Languages

French native
English Proficient

Interests

Sports Fencing (competitor), tennis